How Reliable Is the Estimated Emergency Department Waiting Time?

Maximum likelihood | Bootstrap

Clinical question

An emergency department wants to understand how long non-urgent adult patients (e.g. broken arm but reporting no discomfort) wait between registration and the start of their first assessment by a treating clinician. The team is specifically interested in the population mean waiting time. In a fictional study, 30 independent non-urgent adult patients were each observed for at most five hours after registration. For this exercise, assume that all patients had been assessed at initial triage as non-urgent. For patients whose clinical assessments started during this period, the complete waiting time was recorded. For patients still waiting after five hours, only the fact that their waiting time exceeded five hours was known. Their eventual waiting times are unknown. Assume that no patients left before assessment during the observation period and that there were no other reasons for incomplete observation.

Let the actual waiting time and recorded variables be defined by

$$Y_i = \text{actual waiting time for patient } i, \qquad T_i = \min(Y_i, 5),$$

$$D_i = \begin{cases} 1, & \text{if the clinical assessment starts within five hours,} \\ 0, & \text{if the patient is still waiting after five hours.} \end{cases}$$

A recorded time of five hours with indicator zero is therefore not a completed waiting time of five hours. Example data for the first five patients are given below. Times are measured in hours. The complete dataset EMERGENCY.CSV contains all 30 patients.

Patient	Recorded time	Indicator
1	1.3244898	1
2	0.3768700	1
3	5.0000000	0
4	2.7868014	1
5	0.3691267	1
⋮	⋮	⋮

Your task

Explore the data, construct and fit an exponential waiting-time model that accounts for unfinished waits, and assess uncertainty using observed Fisher information and a nonparametric bootstrap. Present your findings in a clear, informative slide presentation that advises the emergency department's team on the estimated population mean waiting time and its uncertainty. Work in pairs. Use R or Python for the numerical analyses and the statistical theory from the lectures for the derivations.

1. **Explore the data**

   (a) Report the number of patients, the number and percentage with completed waiting times, and the number and percentage still waiting after five hours. Summarize the recorded times using their mean, median, standard deviation, and range.

(b) Make an informative plot showing the recorded times and distinguishing completed from unfinished waits. Choose a display that makes the five-hour observation limit and the incomplete observations clear. Describe the spread and any notable patterns.

2. **Construct the likelihood and derive the MLE**

   Assume that actual waiting times are independent and identically distributed according to an exponential distribution:

$$f_Y(y \mid \lambda) = \lambda \exp(-\lambda y), \qquad y \geq 0, \quad \lambda > 0.$$

   Here, λ is the rate parameter, measured in inverse hours of waiting.

   The following properties of the exponential distribution are relevant for the model:

$$\Pr(Y > c \mid \lambda) = \exp(-\lambda c), \qquad E[Y] = \frac{1}{\lambda}, \qquad \mathrm{Var}(Y) = \frac{1}{\lambda^2}.$$

   (a) Write the likelihood contribution of a patient whose clinical assessment starts before five hours and of a patient who is still waiting after five hours.

   (b) Construct the likelihood and log-likelihood for the full dataset. Derive the score function and maximum likelihood estimator.

   (c) Explain why an unfinished waiting time could provide information about the rate, even though the eventual waiting time is unknown.

3. **Fit the model**

   (a) Calculate the analytical maximum likelihood estimate for the supplied data. Use it to estimate the population mean waiting time.

   (b) Evaluate and plot the log-likelihood over a suitable range of positive rates λ. Mark your analytical maximum likelihood estimate and check whether it agrees with the location of the maximum in your plot. Briefly describe the shape of the curve. Choose and report the range of rates used.

4. **Quantify uncertainty using observed Fisher information**

   (a) Derive the second derivative of the correct log-likelihood and the observed Fisher information evaluated at the maximum likelihood estimate.

   (b) Use the inverse observed Fisher information to obtain an approximate standard error for the estimated rate. Calculate this standard error and an approximate 95% confidence interval using

$$\widehat{\lambda} \pm 1.96\, \widehat{\mathrm{SE}}(\widehat{\lambda}).$$

5. **Quantify uncertainty using the bootstrap**

   Use a nonparametric bootstrap with 2,000 replications.

   (a) In each replication, sample 30 patients with replacement from the observed dataset. Keep each patient's recorded time and assessment indicator together. Calculate the analytical maximum likelihood estimate in every replication. Briefly describe your resampling procedure on your slides.

   (b) Estimate the standard error using the sample standard deviation of the bootstrap rate estimates. Construct a 95% percentile bootstrap interval using their 2.5th and 97.5th percentiles.

(c) Compare the bootstrap standard error and confidence interval for the rate with those from Question 4.

(d) Transform each bootstrap rate estimate into an estimated mean waiting time. Estimate the standard error of the mean waiting-time estimator using the sample standard deviation of these bootstrap estimates, and construct a 95% percentile bootstrap interval for the population mean waiting time. Plot the bootstrap distribution of the mean waiting-time estimates and mark the original estimate and the percentile interval endpoints. No analytical variance formula for this transformation is required.

(e) Explain what increasing the number of bootstrap replications from 2,000 to 20,000 would improve and what it would not improve. Does the bootstrap remove the need for independent, representative observations?

6. **Submission and presentation**

   Submit a **PowerPoint (or LaTeX) presentation of approximately 8 slides**. Include the mathematical steps and a brief description of the bootstrap procedure in the presentation itself. Organize your slides as a coherent story: the clinical question, what was observed, how unfinished waits are handled, the estimated waiting time and its uncertainty, and what the clinical team should conclude. You may use a different allocation, provided that all questions are addressed and the slides remain readable.

   Use descriptive titles, clearly labelled figures, consistent units, and sensible numerical rounding. Wherever possible, use plots and compact tables rather than long paragraphs. Explain the main message of each figure or table. Do not include code or screenshots of raw software output.

Upload your results into the canvas-site before Monday 2 November 2026, 09.00, or email it to m.h.hof@amsterdamumc.nl. Don't forget to write your names in/on the presentation.

Good luck!